

Similar Triangles in Combination

Answer Key

High School Geometry

Quick Answers

- | | |
|-----------------------|---|
| 1. 21 cm | 7. 1.5 |
| 2. 14 in | 8. (a) larger perimeter = 48 cm, — |
| 3. 2.5 | 9. (a) scale factor = 2.5, (b) new perimeter = 210 cm |
| 4. 45 m | 10. (a) area of DEF = 126 cm ² , — |
| 5. 75 cm ² | |
| 6. 17.5 cm | |
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What is verified

8 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 8, 10. The worked solution states what a correct response must contain.

Curriculum

Level: High School Geometry

HSG-SRT.A.2 – *problems 3, 9*

HSG-SRT.B.5 – *problems 1, 2, 4, 5, 6, 7, 8, 10*

Difficulty 1–5 of 5 across 13 tagged checks.

Common wrong answers

- 4. *If they got 67.5: scaled the perimeter by the square of the side ratio*
 - 5. *If they got 45: scaled the area by the side ratio instead of its square*
 - 7. *If they got 2.25: divided the areas without taking the square root*
 - 8. *If they got 76.8: used the area ratio itself as the perimeter ratio*
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1. $\triangle ABC \sim \triangle DEF$ with $AB = 5$ cm, $BC = 7$ cm and $DE = 15$ cm. Find EF , in centimetres.

Solution. $\overline{AB}$ matches $\overline{DE}$ and $\overline{BC}$ matches $\overline{EF}$, so $\frac{5}{15} = \frac{7}{EF}$, giving $EF = 7 \cdot \frac{15}{5}$. Therefore

$EF = 21$ cm

2. $\triangle PQR \sim \triangle STU$ with $PQ = 12$ in, $PR = 21$ in and $ST = 8$ in. Find SU , in inches.

Solution. $\overline{PR}$ matches $\overline{SU}$, so $\frac{12}{8} = \frac{21}{SU}$ and $SU = 21 \cdot \frac{8}{12}$. Therefore

$SU = 14$ in

3. Two similar triangular road signs: the smaller has base 6 cm, the larger has base 15 cm. Find the scale factor from smaller to larger.

Solution. The scale factor is the target length over the starting length:

$$k = \frac{15}{6}.$$

So

$$k = 2.5$$

4. $\triangle ABC \sim \triangle DEF$ with $AB = 8$ m, $DE = 12$ m, and perimeter of $\triangle ABC$ equal to 30 m. Find the perimeter of $\triangle DEF$, in metres.

Solution. From one pair of matching sides, $k = \frac{12}{8} = 1.5$. Perimeter is a sum of lengths, so it scales by the same k , not by k^2 :

$$P_{DEF} = 30 \cdot \frac{12}{8}.$$

Therefore

$$P_{DEF} = 45 \text{ m}$$

5. Two similar tiles have corresponding sides 6 cm and 10 cm; the smaller has area 27 cm^2 . Find the area of the larger, in square centimetres.

Solution. Side ratio $k = \frac{10}{6}$, and area scales by k^2 :

$$A = 27 \left(\frac{10}{6} \right)^2 = 27 \cdot \frac{100}{36}.$$

Therefore

$$A = 75 \text{ cm}^2$$

6. $\triangle ABC \sim \triangle DEF$ with perimeters 24 cm and 60 cm. If $AB = 7$ cm, find DE , in centimetres.

Solution. Perimeters scale by the same factor as sides, so $k = \frac{60}{24} = 2.5$ and

$$DE = 7 \cdot \frac{60}{24}.$$

Therefore

$$DE = 17.5 \text{ cm}$$

7. Two similar triangular plots have areas 48 m^2 and 108 m^2 . Find the ratio of a side of the larger to the matching side of the smaller.

Solution. Areas scale by k^2 , so $k^2 = \frac{108}{48} = 2.25$ and

$$k = \sqrt{\frac{108}{48}}.$$

Therefore

$$k = 1.5$$

8. Two similar sails have areas 50 cm^2 and 128 cm^2 ; the smaller has perimeter 30 cm. (a) Find the perimeter of the larger, in centimetres. (b) Justify why the perimeter ratio is the square root of the area ratio.

Solution (a). The area ratio is $\frac{128}{50} = 2.56 = k^2$, so $k = \sqrt{2.56} = 1.6$ and

$$P = 30\sqrt{\frac{128}{50}}.$$

Therefore

$$P = 48 \text{ cm}$$

Solution (b) — instructor-judged. A complete response says areas of similar figures grow as the *square* of the side ratio, so recovering the side ratio — which is what perimeters follow — means undoing that square with a square root. Using the area ratio directly would scale a length by a factor that belongs to a region. Award full credit only when the squaring relationship is named.

9. A triangular window is enlarged so that its area becomes 6.25 times the original. (a) Find the scale factor applied to each side. (b) The original perimeter is 84 cm; find the new perimeter, in centimetres.

Solution (a). Area scales by k^2 , so $k^2 = 6.25$ and $k = \sqrt{6.25}$, giving

$$k = 2.5$$

Solution (b). Perimeter scales by k , not by k^2 :

$$P = 84\sqrt{6.25} = 84 \cdot 2.5.$$

Therefore

$$P = 210 \text{ cm}$$

10. $\triangle ABC \sim \triangle DEF$ with $AB = 14 \text{ cm}$, $DE = 21 \text{ cm}$, and area of $\triangle ABC$ equal to 56 cm^2 . (a) Find the area of $\triangle DEF$, in square centimetres. (b) Explain why multiplying every side by the same factor multiplies the area by the square of that factor.

Solution (a). $k = \frac{21}{14} = 1.5$, and area scales by $k^2 = 2.25$:

$$A = 56 \left(\frac{21}{14} \right)^2.$$

Therefore

$$A = 126 \text{ cm}^2$$

Solution (b) — instructor-judged. A complete response says a scaled triangle is stretched in two independent directions at once: the base and the height are each multiplied by the factor, and the area, being half their product, picks the factor up twice — once from the base and once from the height. Award full credit only when the response ties the squaring to the two dimensions rather than simply quoting the rule.